

*Exymes Quick-Start Guide*

# **RNA Extraction Using** ***RNAGEM™***



**Exymes**

Ultra Fast RNA Extraction

Find more information at  
**[www.exymesplc.com](http://www.exymesplc.com)**

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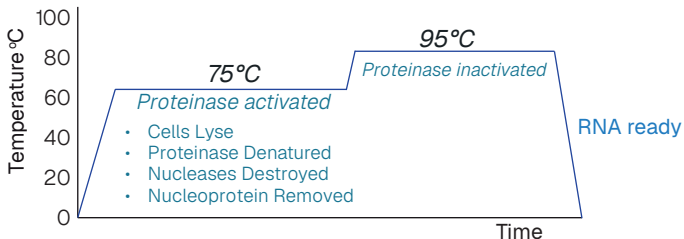
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# RNAGEM™

RNAGEM is for RNA extractions from a range of sample types. More information can be obtained from [www.exymesplc.com](http://www.exymesplc.com)

## Principles of the Extraction

Exymes extraction products use a unique mixture of thermophilic and mesophilic enzymes. Protocols involve a 75°C step to activate Exymes's thermophilic proteinase to lyse cells, denature nucleases, and strip DNA of nucleosomes. A final 95°C step deactivates the thermophilic proteinase resulting in analytics-ready RNA.



## Kit Components and Reagent Storage

RNAGEM reagents are delivered at room temperature but on arrival should be stored at 4°C. After tubes have been opened, the RNAGEM enzyme should be placed at -20°C. Once DNase 1 has been rehydrated, it is stable for 7 months at -20°C. If you do not plan to use all of the DNase 1 immediately, it is recommended that you aliquot DNase 1 into smaller volumes and store at -20°C immediately after rehydration. The buffers can remain at 4°C for convenience.

## Reagent QC

Exymes extraction products use a unique thermophilic proteinase. Extractions are performed at 75°C and the proteinase is inactivated at 95°C. A thermal cycler is used for this process.



## ***General Instructions, Precautions and Technical Tips***

- The method, enzyme formulation and buffer have been carefully optimised for extracting intact RNA. Using the enzyme with other methods or buffers is not recommended.
- If you need to modify the method in any way, please email: [info@exymesplc.com](mailto:info@exymesplc.com) for advice.
- Absorbance 260/280 nm is an ineffective quantitation method with RNAGEM-prepared nucleic acids. For accurate quantification we recommend RT-qPCR and normalisation to genomic DNA using a reference gene, or fluorometric based assays.
- As with any method of RNA preparation, the best results are obtained when samples are handled on ice in an RNase-free environment and using certified RNase-free tubes and reagents.
- For long-term storage, RNA should be stored at -80°C.
- Alternatively, RNA in TE buffer can be precipitated using NH<sub>4</sub>OAc/ ethanol (0.1 volumes of 5 M NH<sub>4</sub>OAc, and 2.5 volumes of 100% ethanol) and stored at -20°C or below
- RNAGEM is sensitive to EDTA and other chelating agents. If cells are presented in EDTA-containing solutions, they should be centrifuged at 2000 x g and washed in 1x **BLUE** buffer before use.
- Cells can be collected directly in the extraction reagent mastermix or the reagents added directly to a capillary from LCM. If cells are collected in a different buffer, it may be necessary to add 1/10th volume of the Exymes buffer after collection. We recommend using Exymes reagents within one hour of preparation. For longer periods, reagents should be frozen.



## Resuspending the DNase I

DNase I is delivered as a lyophilised powder. Before it is ready to use, the powder should be dissolved in 1x DNase I Reaction Buffer (provided as a 10x solution). Different kit sizes contain tubes with different amounts of enzyme (READ THE LABEL). Be sure to add the correct amount of buffer (see the table on the next page).

1. Centrifuge the DNase I tube for 1 minute at 10,000 x g. This will settle the powder to the bottom of the tube.
2. In a clean environment, open the tube and add:

| DNase Rxn Size | 10x DNase Buffer | RNase-Free Water | Total Volume for Reconstitution |
|----------------|------------------|------------------|---------------------------------|
| 100            | 22               | 198              | 220                             |
| 500            | 110              | 990              | 1100                            |
| 1000           | 2x 500 Rxns      | 2x 500 Rxns      | 2x 500 Rxns                     |

*(Exymes supplies extra activity to cater for pipetting error)*

3. Vortex and store at -20°C. The concentration of this solution will be 1 Unit per µl.

*Unit Definition: 1 Unit causes an increase in absorbance at 260nm of 0.001 per minute per ml at 25°C when acting upon highly polymerised DNA at pH 5.0. 0.005 Kunitz unit digests 1 µg of lambda DNA in 10 minutes at 37°C in 50 mM Tris, 1 mM Mg2+, pH 7.8 in a 50 µl reaction.*



## Sample Preparation and Extraction Procedures

*RNAGEM* is a kit for extracting total nucleic acids from mammalian tissue culture and is optimised for producing RNA. The method lyses cells and digests proteins and ribonucleases. Extracted RNA can be used for RT-PCR and RT-qPCR.

All manipulations should be performed in an RNase-free environment or a PCR hood.

Use only certified RNase-free tubes and reagents. *RNAGEM* gives linear yields for 10 to approx  $10^5$  cells and is ideal for single-cell work. For low numbers of cells we recommend reducing the extraction volume. The minimum volume possible will depend on evaporation with the equipment you are using. The recommended amounts of *RNAGEM* to use for different extraction volumes are below. Use 1/10th volume of 10x BLUE buffer.

| Extraction Volume | Cell Numbers     | Volume of <i>RNAGEM</i> |
|-------------------|------------------|-------------------------|
| 50 - 100 µl       | 50,000 - 500,000 | 1 µl                    |
| 20 - 50 µl        | 5,000 - 50,000   | 1 µl                    |
| 5 - 20 µl         | 100 - 5,000      | 0.5 µl                  |
| 1 - 15 µl         | 1 - 500          | 0.2 µl                  |

Sample handling will vary with different sample types. An outline of some suggested procedures is provided on the back page of this document. More information is available at [www.exymesplc.com](http://www.exymesplc.com).



## Handling Different Culture Types

### *Cells in Suspension*

1. Centrifuge the suspension at 200 x g for 5 mins.
2. Remove all of the liquid.
3. Resuspend the pellet in *RNAGEM* extraction reagents.

### *Adherent Cells*

If the cells are in flasks, dislodge cells by preferred method (Trypsin or cell scraper) and centrifuge suspension at 200 x g for 5 mins. Otherwise, the Exymes reagents can be added directly to the adhered layer.

1. Remove all of the liquid.
2. Add *RNAGEM* extraction reagents.

### *Cells Stored in RNeasy<sup>TM</sup>*

1. Centrifuge suspension at 3,000 x g for 5 mins.
2. Remove all of the liquid (a quick spin on a bench centrifuge can help to gather the last few drops).
3. Resuspend the pellet in *RNAGEM* extraction reagents.

### *Cell pellets*

Up to  $5 \times 10^5$  cells can be extracted using the recommended method. Linear extraction efficiency is best achieved within the range of <10 cells to approximately  $10^5$ . Cell pellets can be used directly. Alternatively, the pellet can be resuspended in 1X **BLUE** buffer and an appropriate quantity added to the extraction.

### *Procedure*

#### 1. Add:

Cell suspension or pellet  
5  $\mu$ l 10x **BLUE** Buffer  
1  $\mu$ l *RNAGEM*  
Water to a final volume of 50  $\mu$ l

#### 2. Vortex and incubate:

75°C for 5 min (< 50,000 cells) or 10 min (> 50,000 cells)  
95°C 5 mins (skip if DNA digestion is required)  
4°C HOLD

(A thermal cycler should be used for this step.)



## DNase Treatment (IF REQUIRED)

(Scale for different extraction volumes)

1. To the extract add:
  - 5  $\mu$ l of the 10x DNase buffer
  - 2  $\mu$ l DNase I
2. Vortex and incubate:
  - 37°C for 5 minutes
  - 95°C for 5 minutes
  - 4°C HOLD
3. Add 1/10th volume of 10x TE Buffer (provided) and store at -20°C or below.

## Troubleshooting

For troubleshooting, please see our guide “Exymes Method Optimization and Troubleshooting” in the resources area of the Exymes website.

## Additional Information

For publications and app notes on your specific sample type, please visit our website.



More information can be found at  
[www.exymesplc.com](http://www.exymesplc.com)

If you still need help, email us  
at: [info@exymesplc.com](mailto:info@exymesplc.com)

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